

Multi-Stage Fine-Tuning of Pathology Foundation Models for White Blood Cell Classification

Antony Gitau Martin Paulson Bjorn-Jostein Singstad Karl Thomas Hjelmervik Ola Marius Lysaker Veralia Gabriela Sanchez
University of South-Eastern Norway, Vestfold Hospital Trust
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1. Problem and Challenge Setup

- WBCBench 2026 challenge allowed us to explore how pathology foundation models (FM) perform on a downstream task:
- ▶ **Task:** classify 13 white blood cell (WBC) morphologies from peripheral blood smears.
 - ▶ **Why it is hard:** strong class imbalance, clinically adjacent morphologies, and distribution shift.
 - ▶ **Examples of hard class boundaries:** BNE <-> SNE, MMY <-> MY, PMY <-> MMY, and VLY <-> LY.

Dataset sizes: 33,185 train, 5,350 eval, 16,477 hidden-label test

Split	Largest class	Median class (7th)	Rarest class	Largest / rarest
Train	SNE=15,048	BA=494	PLY=13	1158x
Eval	SNE=2,306	MY=75	PLY=1	2306x

Method Summary

Fine-tuning setup for pathology FM backbone and classifier head on WBC classification as visualized on Figure 2.

- ▶ Three head families share the same backbone: **Linear**, **Cosine**, and **MLP**.
- ▶ End to end training stages S1 -> S2 -> S3 test whether later fine-tuning improves classification.
- ▶ The preprocessing set up is the same for all experiments
- ▶ Inference uses the selected checkpoint with 8-view TTA.

4. Boundary-Dependent Head Behaviour

We conduct a 5-fold out-of-fold (OOF) analysis by training on four folds and evaluating on the held-out fold, then aggregating held-out predictions across all folds.

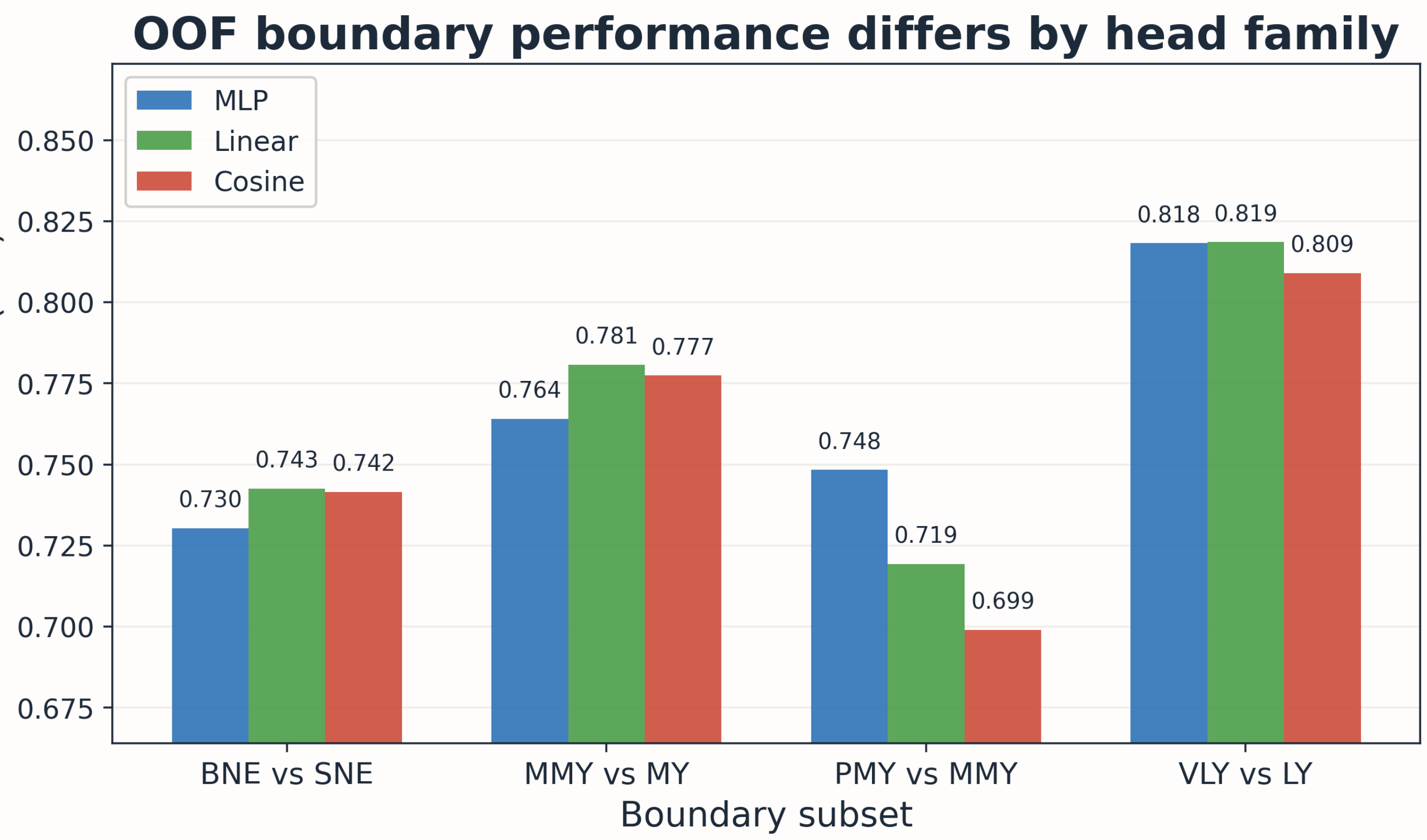


Figure 1: Bars show macro-F1 on four selected hard and morphologically adjacent class pairs, showing that head preference changes across local decision boundaries.

- ▶ BNE <-> SNE: **Linear** and **Cosine** outperform **MLP**.
- ▶ MMY <-> MY: **Linear** is strongest.
- ▶ PMY <-> MMY: **MLP** is strongest.
- ▶ Takeaway from figure 1 is that we observe no single head dominates all hard boundaries, even with the same pretrained backbone underneath.

2. Fine-Tuning and Inference Pipeline

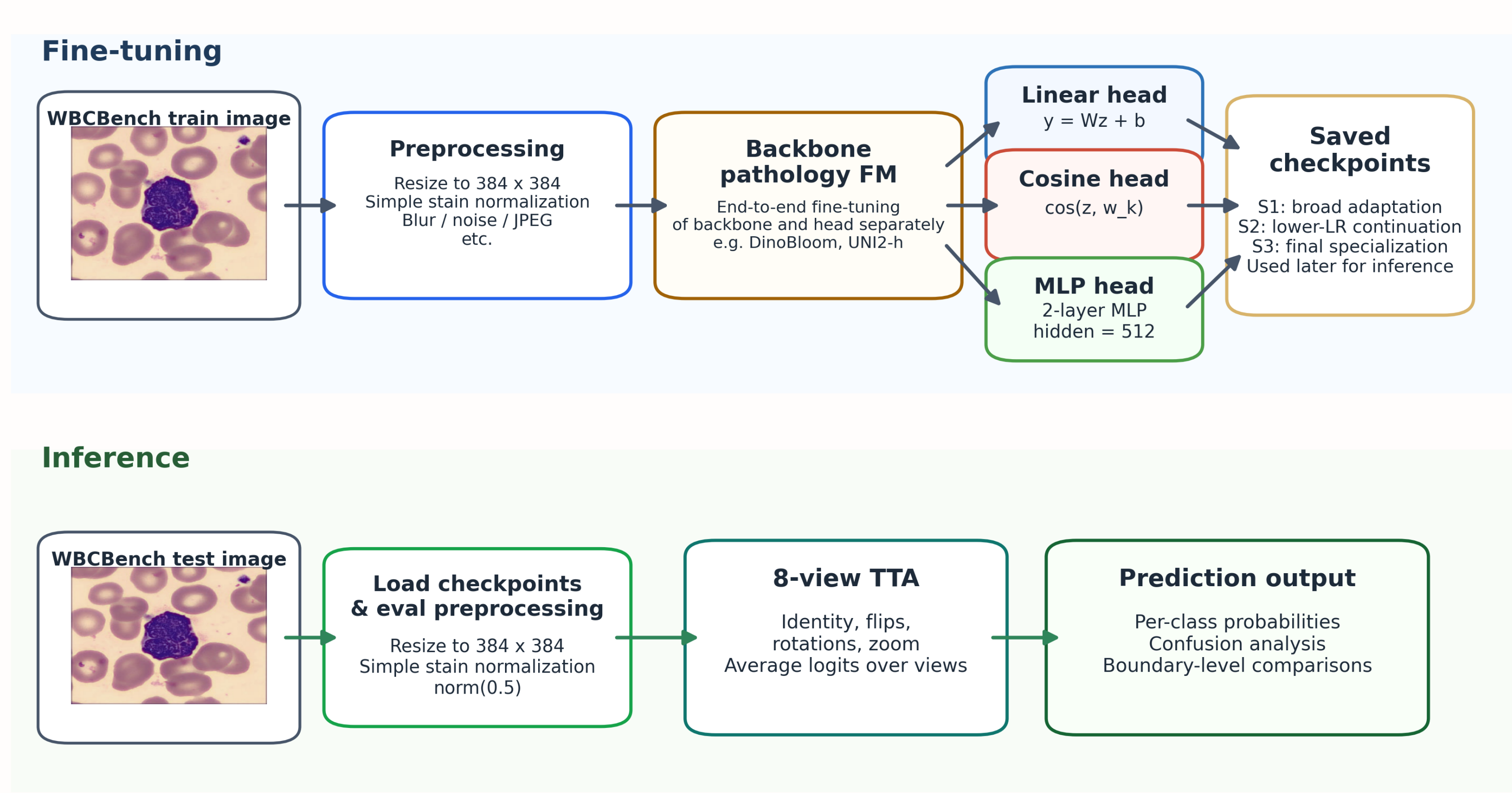


Figure 2: End-to-end fine-tuning and inference simplified visual illustration

A shared pathology foundation model backbone is paired with alternative classifier heads and fine-tuned across three stages (S1→S3), either end-to-end or with head-only updates; resulting checkpoints are then used in an inference pipeline with optional eight-view test-time augmentation.

5. Stage-Wise Eval Behaviour by Head Family

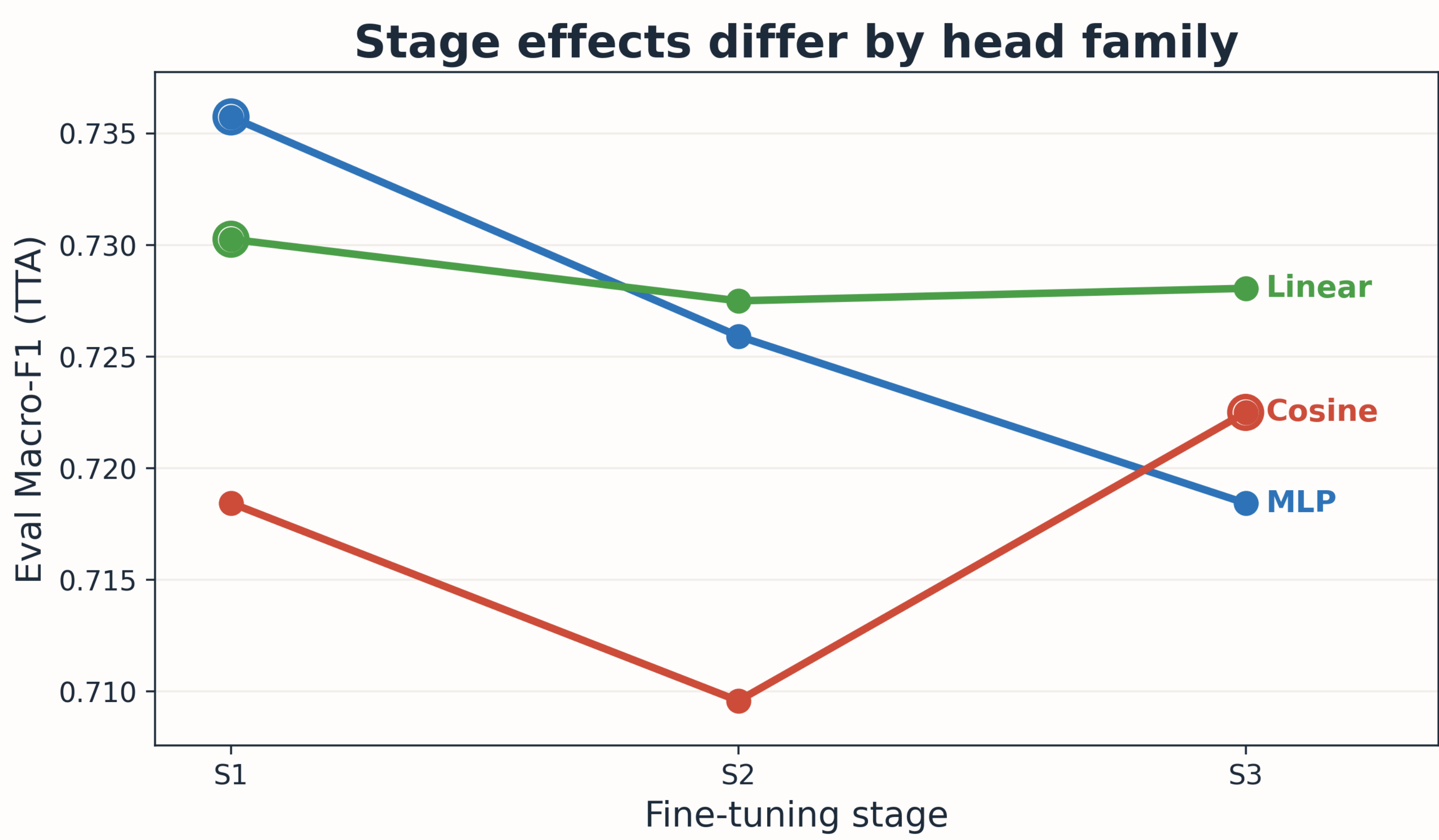


Figure 3: Eval macro-F1 across end-to-end training stages S1 -> S3 for Linear, Cosine, and MLP heads under the same fine-tuning setup visualized in figure 2.

- ▶ **MLP** peaks early at S1.
- ▶ **Cosine** benefits most from later continuation and reaches its best eval score at S3.
- ▶ **Linear** stays competitive across stages, but its best stage is not identical to the other families.
- ▶ Stage choice is part of the head design: one continuation schedule does not fit all heads equally well.

3. Confusion Concentrates on Adjacent Morphologies

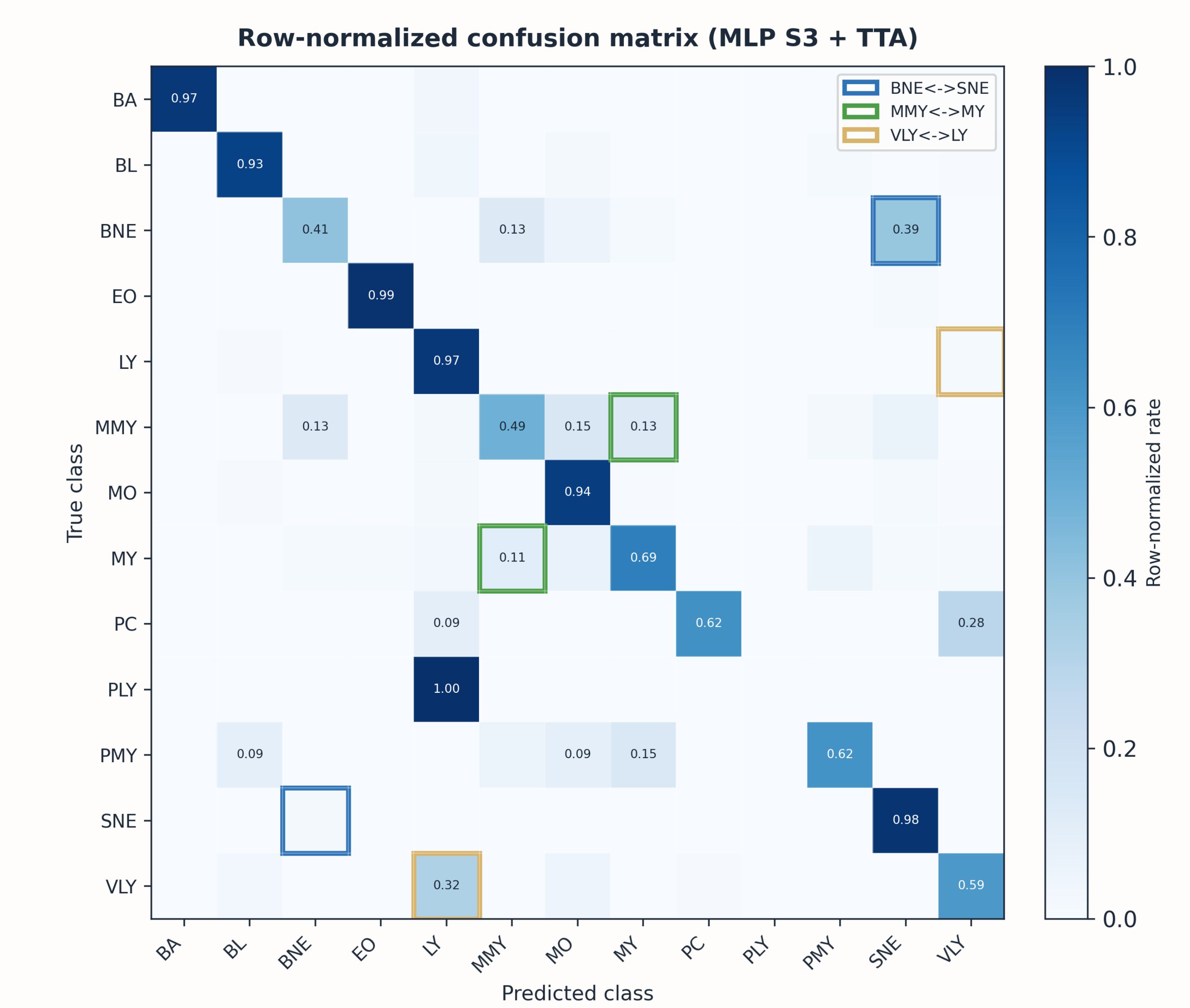


Figure 4: Row-normalized eval confusion matrix for the best single-head checkpoint showing where residual errors concentrate.

- ▶ Even the best single-head checkpoint concentrates errors on a few local morphology boundaries rather than across the whole label space.
- ▶ SNE <-> BNE, MMY <-> MY, and LY <-> VLY remain the dominant structured confusions as highlighted on figure 4.
- ▶ This motivates boundary-level analysis rather than relying only on one global macro-F1 score.

6. Evaluation Lenses

- ▶ **Global view:** eval macro-F1 by stage and head family.
- ▶ **Error view:** row-normalized confusion matrix for the best single-head checkpoint.
- ▶ **Local boundary view:** 5-fold cross-fitted OOF analysis on the hardest morphology subsets.
- ▶ Each OOF prediction comes from a model that did not train on that sample, which gives a cleaner answer than one poster eval split.

Failure Analysis: Naive Tail Rebalancing Can Hurt

Model	BNE recall	BNE precision	BNE F1
Standard MLP	0.521	0.394	0.449
Tail-quota MLP	0.620	0.251	0.358

- ▶ Tail-quota sampling increased BNE recall but pushed the boundary into SNE territory, collapsing precision.
- ▶ Long-tail fixes can backfire when the rare class is morphologically near-identical to a common class.